Virtual Occupancy Sensing from Building CO_2 Data: a Physics-Based Mass Balance versus Machine-Learning Regression

MMA3001 Numerical Methods and Machine Learning — Individual Project
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November 2026

Repository: `github.com/samsamthecodingman/MMA3001_project` (code, HTML documentation in `docs/`, test report in `reports/test_report.html`, decision log `DECISIONS.md`, AI-use statement `AI_USE.md`).

1. Engineering problem

Context. HVAC is typically the largest energy end-use in university buildings. Most systems ventilate to a fixed schedule that assumes design occupancy. *Demand-controlled ventilation* (DCV) instead scales outdoor-air supply to the number of people actually present. It needs a reliable occupancy estimate for every zone. Dedicated people-counters are expensive and are installed in only a few spaces. Low-cost environmental (CO_2/temperature/humidity) sensors are far more common.

Problem addressed. Can the environmental sensors already installed in the Monash Smart Infrastructure (MSI) buildings act as a *virtual occupancy sensor*: estimate the number of occupants in a zone every 15 minutes, validated against the building's people-counters?

Why it matters. A credible virtual sensor would let a facility manager drive ventilation and space-utilisation decisions from existing hardware. To be credible, it needs a quantified error and a known operating envelope.

Intended use. Use the estimates for zone-level ventilation scheduling and utilisation reporting (15-min resolution, errors of ±1–2 people acceptable). They are *not* meant for safety-critical counting such as fire-egress limits.

Scope and its limits. The project compares a physics-based estimator (single-zone CO_2 mass balance), a schedule baseline, and five ML regressors, plus physics/ML hybrids. It does not do online control. Section 2 explains that the usable real data reduced to **one co-located sensor/space pair** (25 days) plus two control pairs. Conclusions about generalisation to other rooms are therefore limited (Section 7).

2. Inputs, outputs and data

2.1 Top-level inputs and outputs

	Quantity	Format / units	Domain
Input	CO_2 concentration	ppm, float, irregular timestamps (~10 min)	350–5000 ppm (else flagged invalid)
Input	Temperature, relative humidity	°C, %	5–45 °C, 0–100 %
Input	Timestamp	tz-aware, Australia/Melbourne	—
Ground truth (training/validation only)	Headcount events from the occupancy counter	persons, integer, event stream (median gap 2–4 s)	0–72 observed
Output	Estimated occupancy per 15-min interval	persons, float ≥ 0	0–~40
Output	Occupied flag	boolean (estimate > 0.5 persons)	—
Output	Calibrated zone parameters (physics model)	effective volume V [m^3], outdoor-air flow Q [m^3/s], supply CO_2 C_out [ppm]	V, Q > 0

Invalid, missing or unsupported inputs.

- Physically implausible readings are set to NaN (ranges in `occupancy.io.VALID_RANGES`).
- Exact re-sent payloads are dropped.

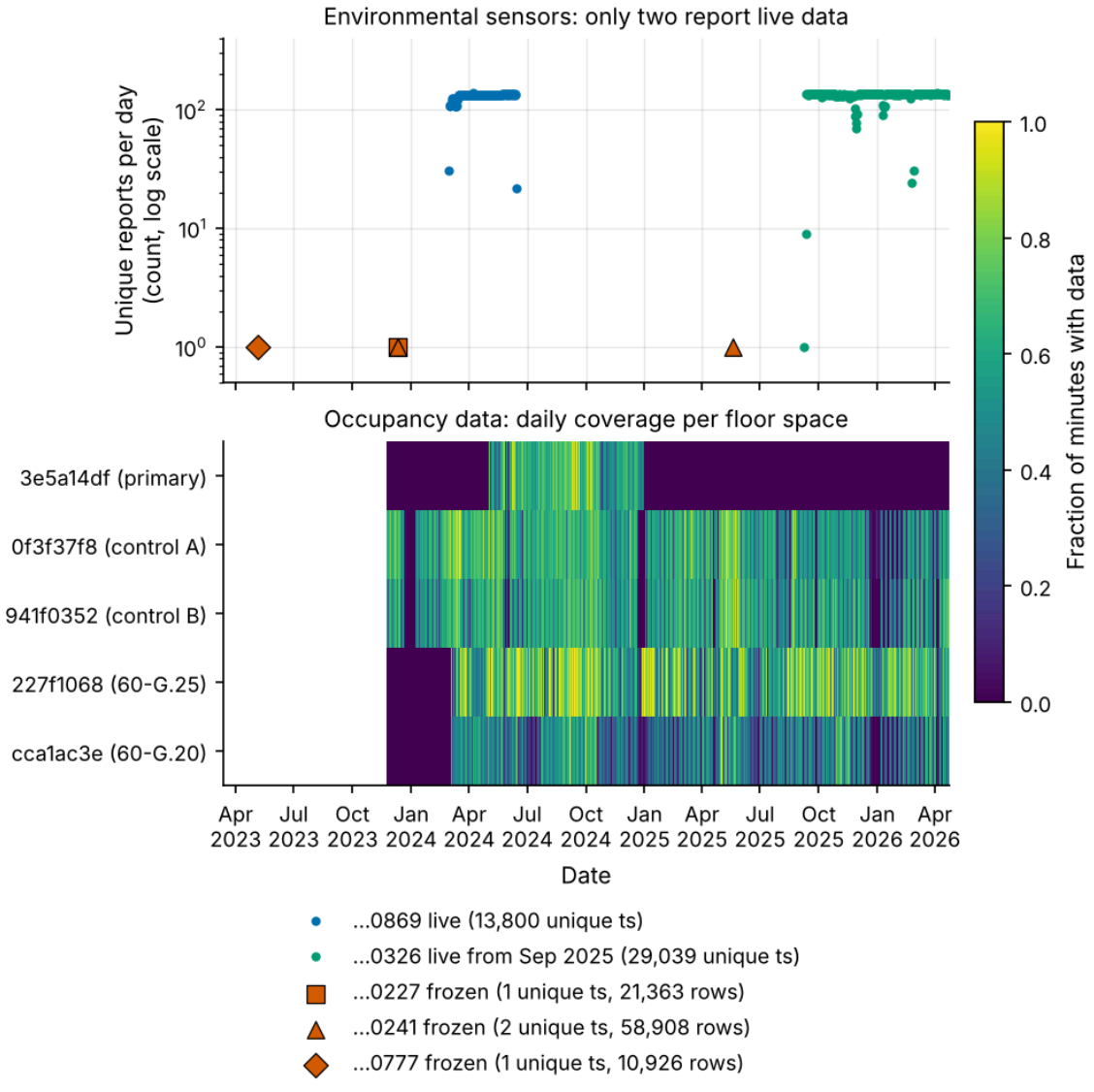
- “Frozen” sensors, with fewer than 100 unique timestamps, are excluded.
- Gaps longer than 45 min are not interpolated, and no estimate is produced inside them.
- Occupancy intervals with less than 50 % counter coverage are dropped. A counter silent for more than 30 min is treated as offline, not as “unchanged”.
- Calibration raises an error if it yields non-physical parameters ($V \leq 0$ or $Q \leq 0$). For example, a room that is never occupied does not excite the dynamics.

Function-level inputs and outputs are documented in the docstrings and the generated HTML documentation (`docs/`).

2.2 Data audit

The teaching team provided two exports: environmental sensors (179 447 rows, JSON payload per row) and occupancy counters (9.09 M event rows, 5 floor-spaces). The audit (`scripts/ingest.py`, Fig. 1) found three problems.

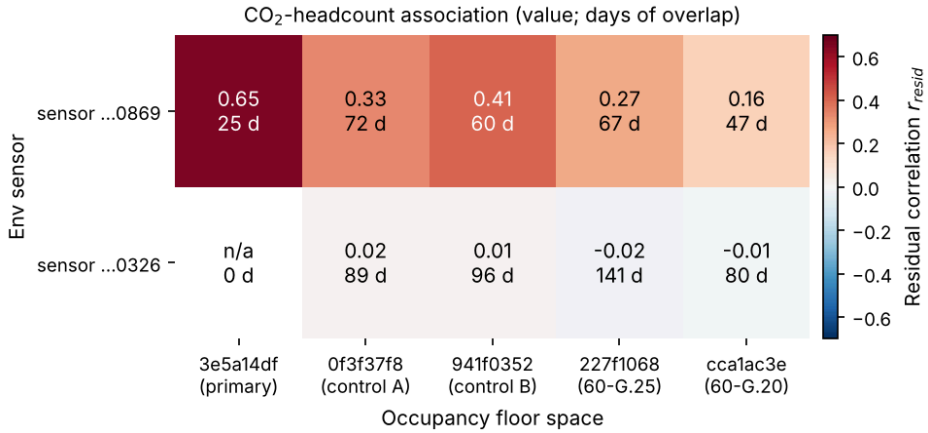
1. **Most env rows are duplicates.** 3 of the 5 env sensors are *frozen*: they re-send a single payload 11 k–59 k times, with 1–2 unique timestamps. In total, 136 604 of 179 447 rows are duplicates. Only two sensors carry live 10-min data: `…0869` (located in G.38, Building 60; Feb–Jun 2024) and `…0326` (not in the location sheet; Sep 2025–Apr 2026).
2. **The location metadata does not link sensors to counters.** The location sheet maps neither live env sensor to a counted space. Its occupancy tab is labelled “SpaceFloorIDs Invalid”, and only 2 of the 5 counter IDs appear in it at all.
3. **The counters are reliable but intermittent.** Counts are integers with plausible ranges. However, there are 1 000–4 700 silences longer than 30 min per space, so the hold cap above is needed.



Data audit: unique env reports per day (top; three sensors are frozen) and occupancy-counter coverage (bottom).

Associating sensors with spaces from data. The metadata could not be trusted, so I correlated each live sensor's CO_2 with each space's headcount on the common 15-min grid. Both series first had their own *time-of-week mean* removed. Without this step, any two series that are high during office hours would correlate.

Sensor `…0869` and space `3e5a14df` had a residual correlation of r = 0.65 over 25 days of labelled data, far above every other pair (Fig. 2). This pair is taken as **co-located (primary)**. Two weaker pairs (r = 0.41, 0.33) are kept as **controls**. Sensor `…0326` correlates with nothing (|r| ≤ 0.03).



Residual (time-of-week-removed) correlation between live env sensors and occupancy spaces.

The "Example Data (Do not use)" files were not used, as instructed by the teaching team.

3. Computational approach

3.1 Pre-processing (interpolation and integration)

Env channels. CO_2, temperature and humidity are continuous fields, so they are *interpolated* onto a uniform 15-min grid. Linear interpolation was chosen by a hold-out test on the real CO_2 record: random observations were withheld (10–50 %) and predicted from the rest.

Method	Hold-out MAE
Linear	7.9–8.4 ppm
PCHIP	about 0.3 ppm worse than linear
Nearest-neighbour	9.3–9.8 ppm
Natural cubic spline	10.0–11.2 ppm (overshoots on noisy data)

Gaps longer than 45 min are left empty.

Headcount. Headcount is a *state reported at events*, so it must be averaged over each interval, not interpolated. The 15-min mean is the time integral of the zero-order-hold signal divided by the covered time.

A first implementation looped over events in Python. The final version (`io.occupancy_on_grid`) evaluates a cumulative integral at the grid edges in vectorised NumPy:

$$I(x) = S_k + w_k h_k (x - t_k), \qquad S_k = \sum_{j<k} w_j h_j d_j .$$

This aggregates 1.5–2.2 M events per space in 0.1–0.4 s. Both versions agree to 1e-9 on random event streams and on a hand-calculated case (`tests/test_io.py`).

Target alignment. CO_2 is an instantaneous sample, whereas the target is an interval mean. Labelling the mean over $[t, t+\Delta t)$ at t makes occupancy *lead* CO_2 by Δt/2. The synthetic study (§4) showed this biases the calibrated volume by +31–52 % at Δt = 15 min. The target is therefore **centred**, $[t-\Delta t/2,\ t+\Delta t/2)$, which reduced the bias to +2–19 %.

3.2 Physics-based estimator (ODE, finite differences, least squares)

The zone is modelled as a single well-mixed control volume:

$$V\frac{dC}{dt} = 10^6\, g N(t) - Q\,(C - C_{\text{out}}).$$

Here $g = 5.2 \times 10^{-6}$ m³/s per person is the CO_2 generation of a sedentary adult [1, 2].

- **Forward problem (verification and simulation).** Three solvers are implemented: explicit Euler, classical RK4, and the exact solution for piecewise-constant N. On a uniform grid, the exact update $C_{k+1} = aC_k + (1-a)C_{ss,k}$, with $a = e^{-\Delta t/\tau}$ and $\tau = V/Q$, is a first-order IIR filter evaluated with `scipy.signal.lfilter`.
- **Inverse problem (the estimator).** $\hat{N} = (V\dot{C} + Q(C - C_{\text{out}}))/(10^6 g)$. The derivative $\dot{C}$ comes from a finite-difference scheme: backward (1st order), central (2nd order) or Savitzky–Golay (local quadratic fit, window 5), applied run-by-run so gaps do not contaminate neighbours.
- **Calibration.** The model is linear in (V, Q, QC_{out}), so $N = a\dot{C} + bC + c$ is solved by ordinary least squares on training data. This gives $V = 10^6 ga$, $Q = 10^6 gb$ and $C_{\text{out}} = -c/b$. Calibration is one $n \times 3$ least-squares solve: closed form, unique, and fast (2 ms).
- **Assumptions.** Perfect mixing; constant Q and C_out within the calibration window; constant per-person generation; no other CO_2 sources or sinks; the sensor represents the zone.

3.3 Data-driven and hybrid models

All models are scikit-learn regressors on **near-causal** features (current and past grid values only; the linear interpolation onto the grid looks at most one raw sample, about 10 min, ahead), so they could run in near real time:

- CO_2 and its first difference at lags 0–3;
- CO_2 above a rolling 24-h 5th-percentile baseline;
- temperature, humidity and their differences;
- hour of day (sin/cos);
- weekend flag.

Model	Configuration
Ridge linear regression	Linear, regularised
Decision tree	depth 8, min-leaf 20
Random forest	200 trees, depth 12, min-leaf 10
SVR	RBF kernel
MLP	64-32 ReLU, L2 = 1e-3, Adam, early stopping

Baseline. The *time-of-week mean*, which uses no sensor data, sets the bar any sensor-based model must clear.

Hybrids. The forest and MLP are also trained with the physics estimate as an extra feature.

3.4 Validation design

Main protocol: blocked leave-one-week-out cross-validation. Each calendar week is predicted by models trained on all *other* weeks. A 1-day buffer either side stops lagged and rolling features, and autocorrelation, from leaking across the boundary. Out-of-fold predictions are pooled and scored once.

Why not other splits.

- Random K-fold splitting would leak information between neighbouring samples.
- A purely chronological split was also run (60/20/20). Its 8-day test window fell entirely in the June exam period, where mean occupancy was 0.8 people against 3.7 in training. R^2 is meaningless on such low test variance. That split is therefore reported separately as a regime-shift scenario (§5.3).

Metrics.

- MAE in persons, the operator-relevant error;
- RMSE;
- R^2;
- bias;
- **balanced accuracy** for the occupied/vacant state (> 0.5 persons).

F1 and plain accuracy are *not* used as evidence of skill. 77 % of labelled intervals are occupied, so a constant “occupied” prediction already scores F1 = 0.87. Balanced accuracy gives 0.5 for any constant prediction. All models are scored on the common set of intervals where every model has a prediction.

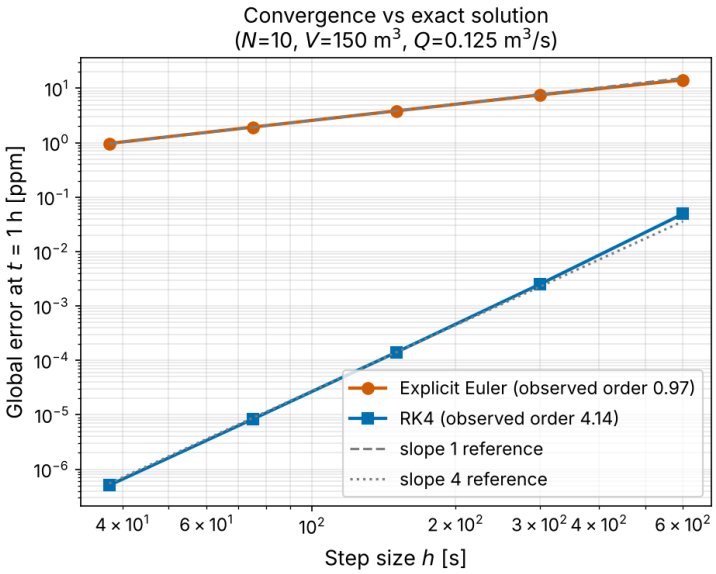
The validation folds did not influence the headline model settings. The defaults were fixed beforehand, and tuning is reported only as sensitivity (§6.2) to avoid optimistic selection bias.

4. Verification (does the code solve the equations correctly?)

Verification uses manufactured data where the truth is known exactly (`tests/`, `scripts/sensitivity_synthetic.py`).

- **Order of accuracy (Fig. 3).** Against the analytical solution, the observed global orders are 0.97 for Euler and 4.14 for RK4 (theory: 1 and 4).
- **Stability (Fig. A0).** Explicit Euler on $\dot{C} = -C/\tau + f$ has amplification factor $|1 - \Delta t/\tau|$. It oscillates for $\Delta t > \tau$ and diverges for $\Delta t > 2\tau$, exactly as predicted. Typical rooms have $\tau \approx 20$–60 min, so a 15-min explicit step is near the oscillation limit for small rooms. This is one reason the exact solution is used.
- **Inverse and calibration.** On noise-free data, the inverse recovers constant occupancy to ±0.05 persons. Calibration recovers V and Q within 5 % and C_out within 5 ppm.
- **End-to-end on synthetic rooms.** Four rooms were generated with irregular ~10-min sampling, 15 ppm noise and 3 % dropouts. Physics R^2 was 0.96, and the calibrated V and Q were within 4–17 % and 5 % of truth respectively.
- **Vectorisation.** The vectorised solver matches the loop to 1e-10.

All 27 unit tests pass (`reports/test_report.html`).



Convergence of explicit Euler and RK4 against the exact solution (observed orders 0.97 and 4.14). The stability demonstration is in Appendix A (Fig. A0).

5. Validation on real data

5.1 Main result (blocked CV, primary pair, 7 folds, 2 414 intervals)

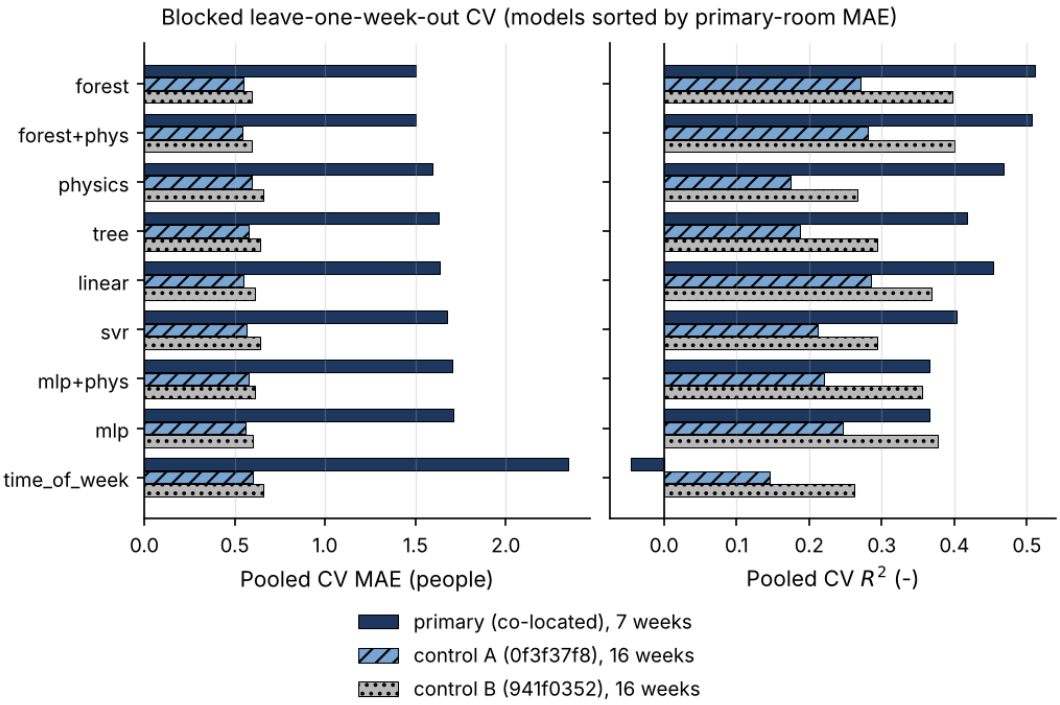
Model	MAE [persons]	RMSE	R^2	Bias	Balanced acc. (occupied)	Fit time per fold
Time-of-week baseline	2.35	3.21	−0.05	+0.09	0.52	2 ms
Physics (3 parameters)	1.60	2.29	0.47	**+0.03**	0.56	**2 ms**
Linear (ridge)	1.63	2.32	0.45	+0.17	**0.65**	3 ms
SVR	1.68	2.42	0.40	−0.01	0.61	0.23 s
MLP 64-32	1.71	2.50	0.37	+0.16	0.57	0.69 s
Random forest	**1.50**	**2.19**	**0.51**	+0.09	0.55	0.56 s
Forest + physics feature	1.50	2.20	0.51	+0.09	0.56	0.55 s

The sensor carries real information. Every sensor-based model beats the schedule baseline. MAE falls by 0.64–0.84 persons (−27 % to −36 %), and R^2 rises from −0.05 to 0.37–0.51. On the two *control* pairs (sensor not co-located), the same forest reaches only R^2 = 0.27/0.40, against 0.15/0.26 for the baseline (Fig. 4). The gain is far smaller, which supports the claim that the primary pair measures room-level occupancy rather than building-wide activity.

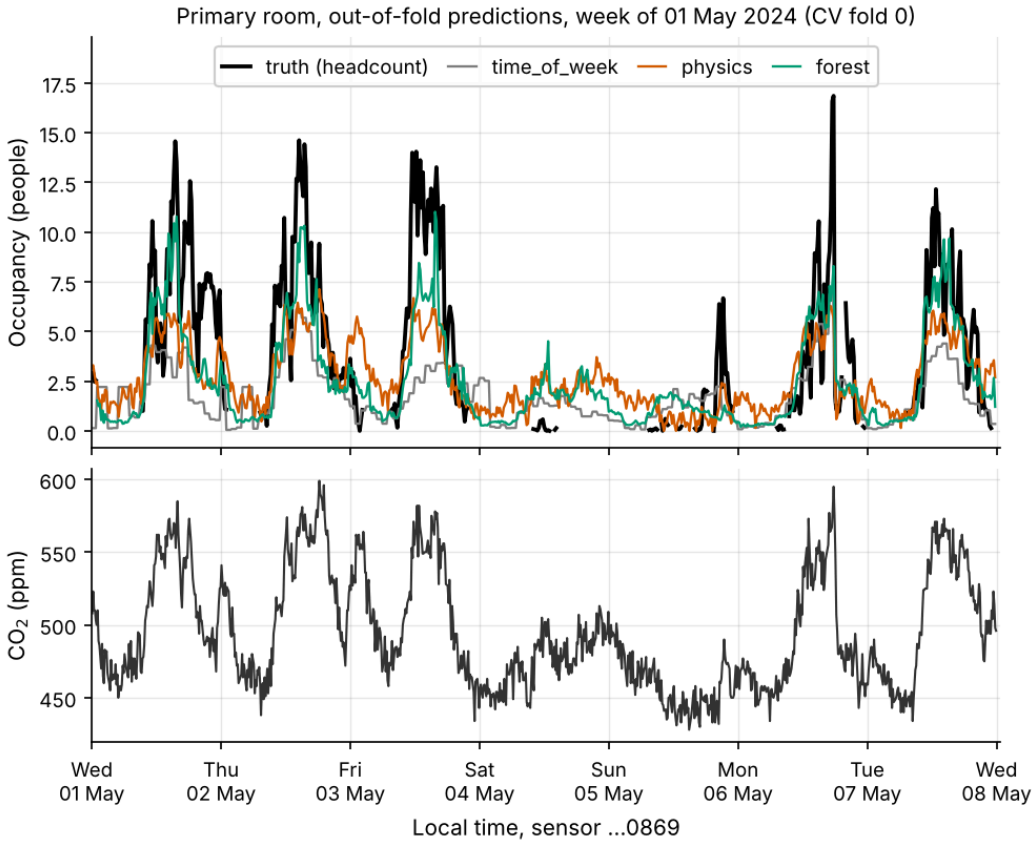
Physics versus black box. The 3-parameter physics model is within 0.09 persons MAE of the best model. It has the smallest bias, and it fits 300× faster than the forest. The MLP is not better than linear regression on 2 600 samples of tabular data. Adding the physics estimate as a forest feature changed nothing, unlike on synthetic data, where it helped (MAE 0.15 → 0.10). On real data the forest already extracts the same CO_2-level information.

Vacancy detection is weak. Balanced accuracy for occupied/vacant is only 0.52–0.65 (linear regression is best). The models estimate *how many* people are present far better than *whether anyone* is. After people leave, CO_2 decays over tens of minutes, so recently vacated intervals still look occupied. A virtual sensor of this kind should not be used for vacancy-triggered set-backs without a hysteresis or time-out rule.

Failure mode. All models **under-predict peaks above ~10 people** (Fig. 5). The CO_2 at a sensor mounted high in a large workshop zone rises only 100–150 ppm for 15 occupants, because the effective volume is large. Accuracy at high occupancy is therefore limited by physics, not by the model class.



Blocked-CV MAE and R^2 for the primary and control pairs.



Out-of-fold predictions for a teaching week (1–7 May 2024), with the measured CO_2 below.

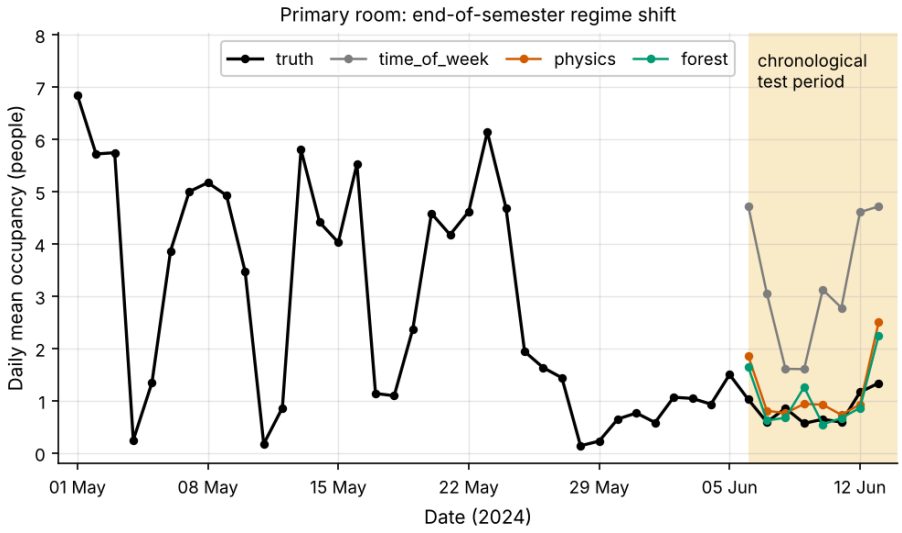
5.2 What the validation cannot establish

- **Single site.** There is only one co-located pair, with 25 days in one season. Accuracy in other rooms, other ventilation regimes (e.g. winter heating), or with occupancy far outside 0–17 is untested.
- **Uncertain pairing.** The co-location is *inferred* statistically, not confirmed physically.
- **Imperfect ground truth.** The counter itself is not error-free. Its own error is unknown and is included in the reported MAE.

- **Overlapping folds.** CV folds share the same 6-week period, so the result is not a test of long-term drift. Leave-one-week-out also trains on *later* weeks, which is optimistic compared with forward deployment (§5.3 covers the forward case).
- **Selection optimism.** The primary pair was selected as the best of 10 sensor–space pairs, using correlation over the same labels that CV later scores. The primary-pair R^2 is therefore optimistically biased. The controls only partly offset this; selecting on an independent period would be the rigorous fix.
- **Label availability is not uniform.** Counter silences longer than 30 min are treated as missing, not as zero. Night-time label coverage is only 17–33 %, against 80–93 % by day, so the scored intervals over-represent busy periods.
- **Coverage floor.** A 15-min interval is scored only if at least 50 % of its minutes have counter data.

5.3 Regime shift (chronological split)

When trained on May (teaching) and tested on 6–13 June (exams), the schedule baseline fails badly: MAE 2.42, because it predicts teaching-week occupancy. The CO_2-based models adapt (forest MAE 0.65, physics 0.85; Fig. 6). This is the practical argument for a sensor-based estimate over a timetable. A timetable cannot know that the semester has ended.



Daily mean occupancy over the paired period; the shaded chronological test window falls in the exam period.

6. Sensitivity, optimisation and performance

6.1 Physics-model sensitivity

Derivative scheme and grid spacing (Fig. 7a).

- Prediction accuracy is almost insensitive to the derivative scheme (CV R^2 0.45–0.49 across backward, central and Savitzky–Golay) and improves slightly at a 30-min grid.
- The backward-difference result at 15 min looks best, but is scored on only 5 of 7 folds; the other two calibrations were non-physical.

Parameter identifiability. The *identified* volume is not robust. V ranges from 23 to 545 m^3 across schemes, whereas Q stays at ≈ 0.25 m^3/s (≈ 250 L/s). The estimate is dominated by the steady-state term $Q(C - C_{out})$. With 10-min sampling and transients shorter than τ, the derivative term is poorly identified. Engineering consequence: the calibrated Q is a credible ventilation estimate, but V should not be read as a physical room volume.

Generation rate g. It follows algebraically from the linear calibration that predictions are invariant to the assumed g; numerically, they were identical to machine precision for g = 3.5–7.0 × 10^{-6} m^3/s. Only the inferred V and Q scale with g. An uncertain metabolic rate therefore does not affect the occupancy estimate when the model is calibrated against counts.

Synthetic sensitivity.

- Sensor noise from 0 to 50 ppm raised MAE from 0.10 to 0.60 persons.

- Noise moved the V error from +23 % to −46 % and the Q error from 0 to −20 %. This is the classic errors-in-variables attenuation of least squares when the regressor (the noisy derivative) is itself measured with error.

6.2 Model selection and data requirement

Hyper-parameter sweep.

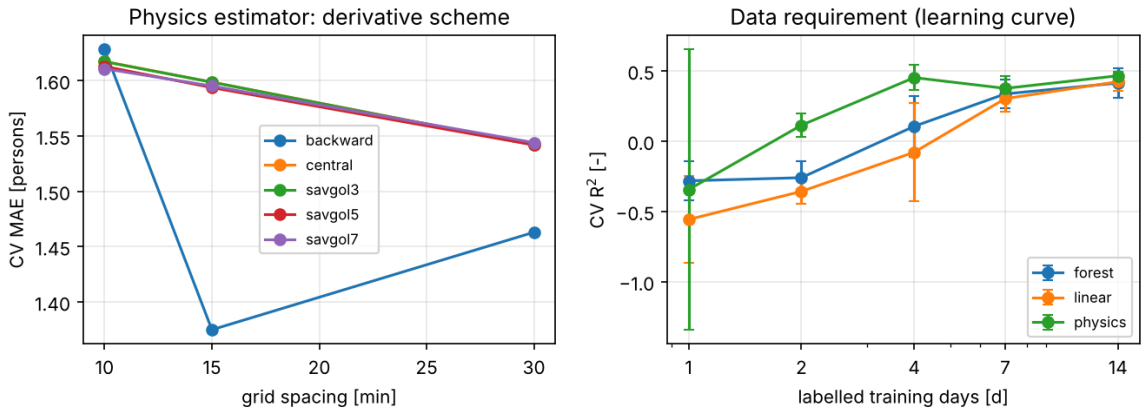
- Forest depth 4–∞ and leaf size 5–50: R^2 0.504–0.536.
- MLP width 16–64-32 and L2 1e-4–1: R^2 0.34–0.40.

The best forest is only 0.02 above the default, and it was selected on the same folds, so the defaults were kept.

Learning curve (Fig. 7b). Each fold's training set was subsampled to *n* labelled days.

- The physics model reaches $R^2 = 0.45 \pm 0.09$ with **4 days** of labels.
- The forest needs about 14 days to reach a similar level (0.11 at 4 days, 0.41 at 14).

For commissioning a new room, the physics model's data efficiency matters more than the forest's final 0.05 R^2.



(a) Physics estimator MAE vs grid spacing and derivative scheme. (b) Learning curve: CV R^2 vs labelled training days.

6.3 Computational performance

Machine. 4 vCPU Xeon @ 2.1 GHz with AVX-512 (2 FMA units).

- Theoretical peak: $F_T = 4 \times 2.1\,\text{GHz} \times (512/64) \times 4 = 269$ GFLOP/s.
- Measured dgemm: 188 GFLOP/s.
- Measured stream bandwidth: 8.4 GB/s.
- Machine balance ≈ 22 FLOP/byte.

Method	FLOP / sample	Bytes / sample	Arithmetic intensity	Measured time (n = 10^6)
RK4 step (Python loop)	32	24	1.3	2 000 ns
Exact step, Python loop	6	24	0.25	295 ns
Exact step, `lfilter`	6	24	0.25	**7.5 ns**
Inverse estimator (central)	8	16	0.5	8.3 ns
Inverse estimator (Sav.–Golay)	15	16	0.9	21 ns
MLP inference (24-64-32-1)	7 300	200	37	—

Physics estimator. Its intensity (0.5–0.9 FLOP/B) is far below machine balance, so it is **memory-bound**. The roofline bound is about 2 ns/sample (16 B ÷ 8.4 GB/s). The measured 8.3 ns reflects NumPy temporaries, i.e. several passes over memory.

MLP. It is compute-bound when batched. It needs ~500× more FLOP per estimate than the physics model.

Optimisation results.

- Replacing the Python time-stepping loop with the IIR-filter form of the exact solution gave a **40× speed-up** over the exact loop, and about 260× over the RK4 loop, with identical results (Fig. A1).
- Vectorising the occupancy aggregation turned a per-event Python loop into a cumulative-sum evaluation (0.1–0.4 s per 2 M events).
- Profiling with cProfile shows that the full experiment time is dominated by SVR and random-forest fitting inside CV, not by the numerical methods.

At building scale (hundreds of zones at 15-min resolution), every method is computationally trivial. The deciding criteria are therefore accuracy, data requirement and interpretability, not cost.

7. Limitations and lessons learned

Limitations.

1. **Narrow evidence base.** Real-data evidence rests on one inferred co-located pair over 25 days. This is the largest threat to external validity.
2. **Single-zone assumption.** The model assumes one well-mixed zone. A sensor mounted high in a large workshop sees a diluted, delayed signal, which is why every model under-predicts peaks.
3. **Latency.** The Savitzky–Golay derivative adds 30 min of latency.
4. **No uncertainty estimate.** No prediction intervals are produced, even though DCV control should use them.

Lessons.

1. **Data auditing changed the project.** 76 % of env rows were duplicates, and the metadata was wrong. The physical pairing of sensor and space had to be inferred from data, with controls to test that inference.
2. **Evaluation design mattered more than model choice.** A naive chronological split produced R^2 of −23 for the baseline and negative values for every model. The cause was an exam-period test window, not bad models. Blocked CV gave a fair picture.
3. **A small timing detail produced a large parameter bias.** Labelling the interval mean at its start biased V by +40 %; centring fixed it. Synthetic verification exposed the bug before real data could hide it.
4. **Simple models are hard to beat.** A 3-parameter physics model with one least-squares solve matched an MLP. It needs a quarter of the training data, and its parameter Q has engineering meaning.

Proposed improvements, in priority order.

1. Confirm the sensor–space pairing on site, and obtain co-located sensors for more rooms, to enable leave-one-room-out validation.
2. Add a two-zone or delayed-mixing model.
3. Fit an output-error (simulate-and-match) calibration with a Kalman-filter state estimator, giving uncertainty bounds and less noise sensitivity.
4. Use the physics model for commissioning and switch to a forest once more than 2 weeks of labels exist.

8. AI use (summary)

Full statement in `AI_USE.md`. AI tools (Anthropic Claude, via Claude Code) were used extensively, as the unit permits:

- planning;
- writing most of the code and tests;
- running analyses;
- drafting this report.

The student set the goal and constraints, selected the dataset, supplied the data, and is responsible for reviewing every decision in `DECISIONS.md` and every claim in this report. My role was to orchestrate rather than to write code: I split the work into stages with acceptance criteria (numerics verified on synthetic rooms before real data; every model must beat a no-sensor baseline; validation must respect time order), reviewed each stage's evidence before approving the next, and commissioned an independent AI review.

Most errors were caught by the AI's own tests and diagnostics (see `AI_USE.md` §5); my main intervention was commissioning a separate reviewer agent so the code was not only checked by the agent that wrote it, which exposed a misleading F1 score that was then replaced by balanced accuracy. The trade-off is a shallower grasp of low-level implementation, which I offset by working through the decision log so I can justify each choice.

References

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[2] A. Persily and L. de Jonge, "Carbon dioxide generation rates for building occupants," *Indoor Air*, 27(5), 868–879, 2017.

[3] Monash Smart Infrastructure, *MSI occupancy and environmental sensor exports (May–Dec 2024)*, provided for MMA3001, Monash University, 2026.

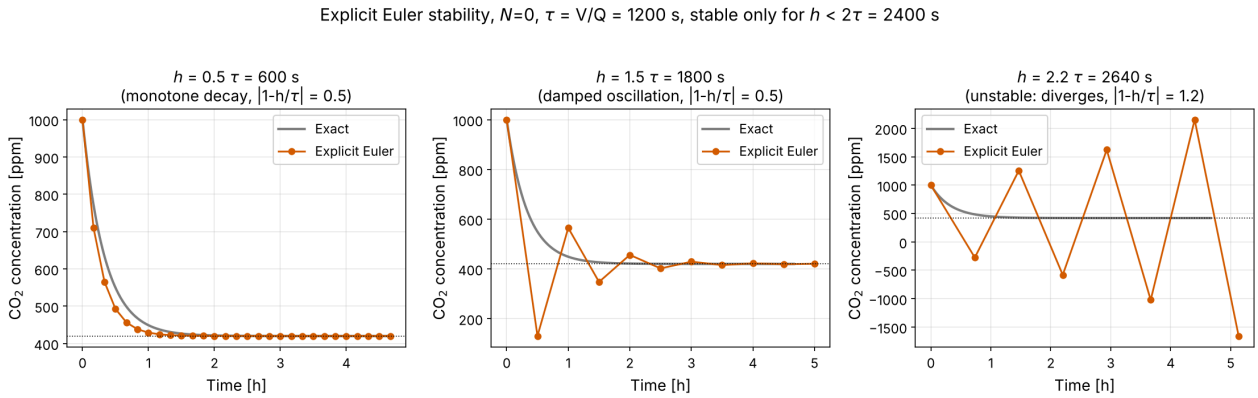
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[5] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *JMLR*, 12, 2825–2830, 2011.

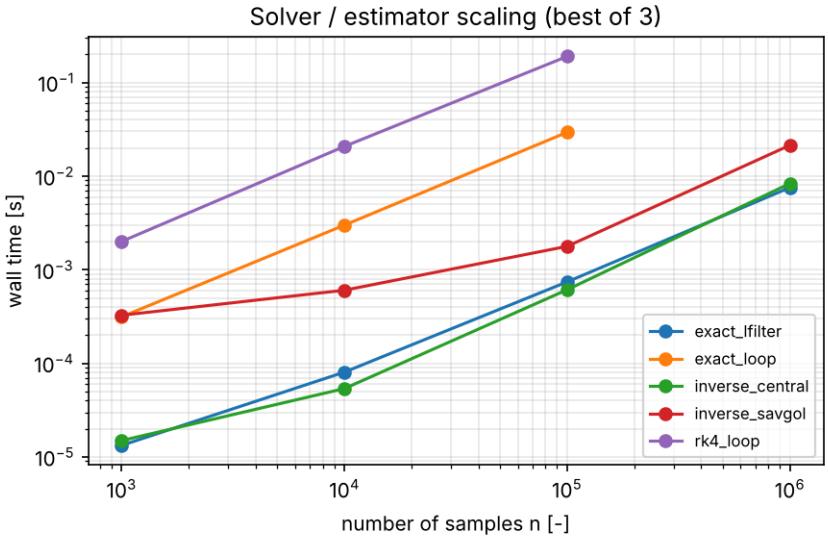
[6] A. Savitzky and M. J. E. Golay, "Smoothing and differentiation of data by simplified least squares procedures," *Analytical Chemistry*, 36(8), 1627–1639, 1964.

[7] MMA3001 unit notes, weeks 2–11 (optimisation and profiling, interpolation, ML regression, finite differences, ODE stability), Monash University, 2026.

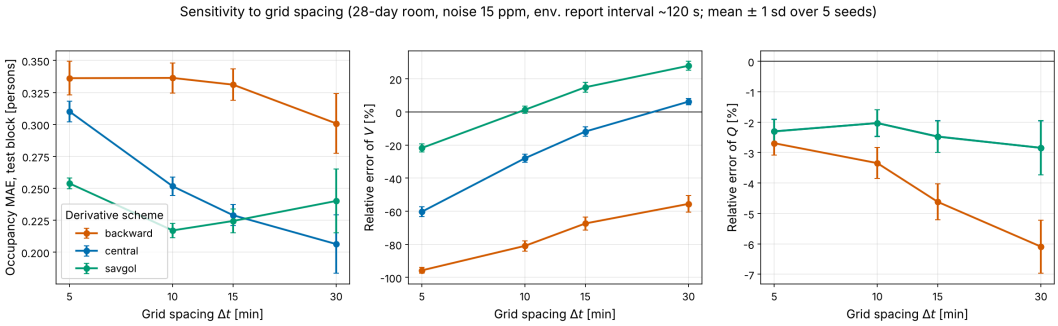
Appendix A — additional figures



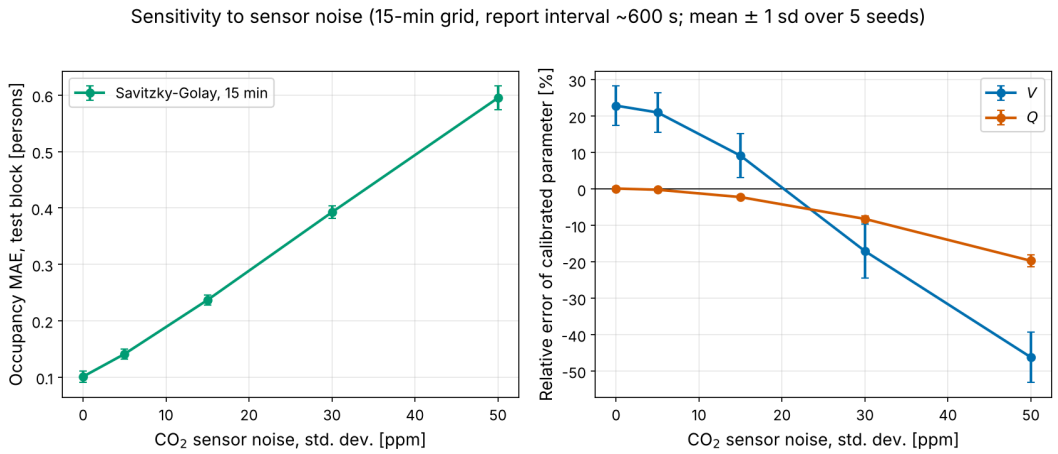
A0: Explicit-Euler decay from 1000 ppm with N = 0 for Δt = 0.5τ (stable), 1.5τ (oscillatory) and 2.2τ (divergent), against the exact solution.



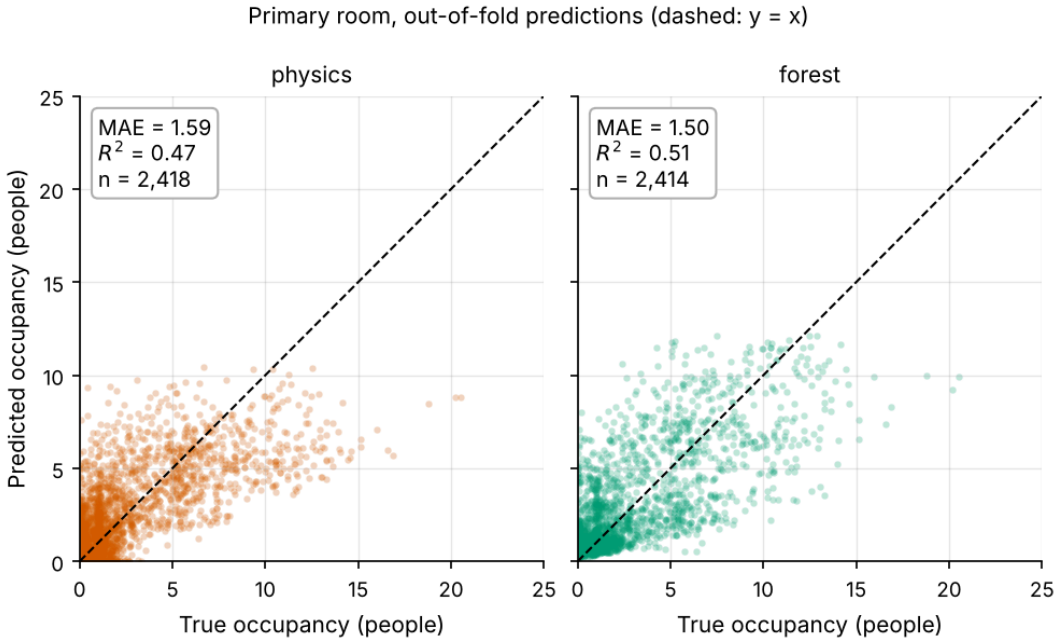
A1: Wall time vs problem size for solvers and estimators.



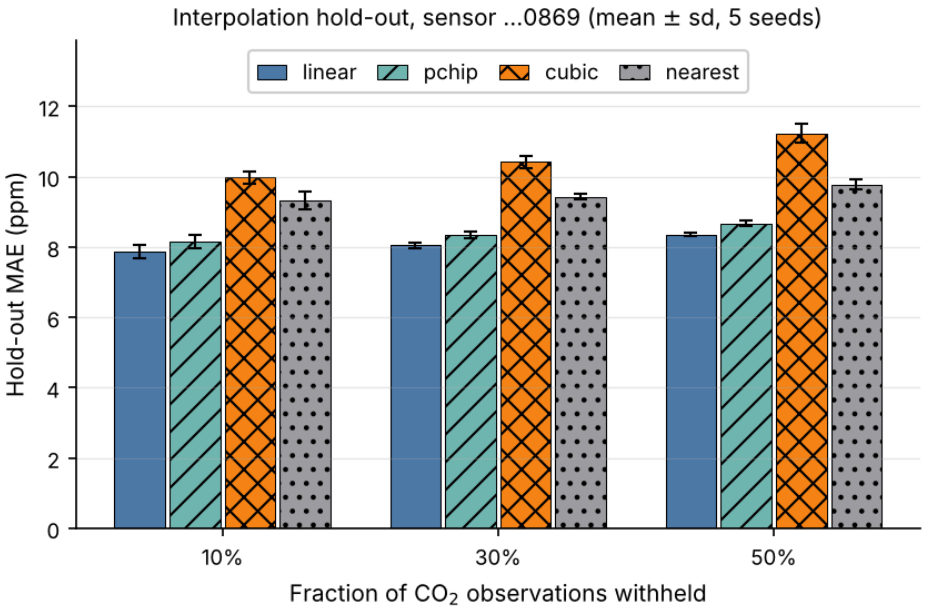
A2: Synthetic sensitivity of the physics estimator to grid spacing and derivative scheme.



A3: Synthetic sensitivity to CO_2 sensor noise.



A4: Predicted vs true occupancy (primary pair, out-of-fold).



A5: Interpolation hold-out error on real CO_2.